

● HARD

● ADVANCED MATH

Equivalent expressions

9 questions · ~14 min

03

Q1

GRID-IN ADVANCED MATH

If $4^{8c} = \sqrt[3]{4^7}$, what is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In the expression $3(2x^2 + px + 8) - 16x(p + 4)$, p is a constant. This expression is equivalent to the expression $6x^2 - 155x + 24$. What is the value of p ?

- A. -3
- B. 7
- C. 13
- D. 155

$$9x^4 + 8x^3 - 7x^3 + 5x^2 - 3x^2 + x$$

The given expression can be written in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. What is the value of a ?

- A. -8
- B. -2
- C. 2
- D. 8

$$\frac{x^2 - c}{x - b}$$

In the expression above, b and c are positive integers. If the expression is equivalent to $x + b$ and $x \neq b$, which of the following could be the value of c ?

- A. 4
- B. 6
- C. 8
- D. 10

If a and c are positive numbers, which of the following is equivalent to

$$\sqrt{(a+c)^3} \cdot \sqrt{a+c}?$$

- A. $a+c$
- B. a^2+c^2
- C. $a^2+2ac+c^2$
- D. a^2c^2

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A. $3x^2 + 7x + 14b$
- B. $3x^2 + 28x + 14b$
- C. $3x^2 + 42x + 14b$
- D. $3x^2 + 49x + 14b$

Q7

GRID-IN ADVANCED MATH

The expression $3x - 2319x + 6$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

- A. $(x + a)^2$
- B. $(x - a)^2$
- C. $(x + a)(x - a)$
- D. $x^2 - ax - a^2$

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $hx + kx + j$, where h , k , and j are integer constants. Which of the following must be an integer?

- A. $\frac{b}{h}$
- B. $\frac{b}{k}$
- C. $\frac{45}{h}$
- D. $\frac{45}{k}$